Squeezing-enhanced Key Rates in Continuous Variable Quantum Key Distribution under Collective Beamsplitter Attack and Imperfect Reconciliation

Konstantinos Pourgourides

Faculty of Applied Sciences, Delft University of Technology

Internship Supervisor

Professor Ioannis Krikidis

IRIDA Research Center for Communication Technologies, University of Cyprus

Summer 2026

Nicosia, Cyprus

Abstract

Quantum Key Distribution (QKD) refers to the implementation of a quantum cryptographic protocol which aims to produce a secret shared key between two parties (Alice and Bob) in the presence of an eavesdropper (Eve). Contrary to Classical Key Distribution (CKD), the security of QKD is embedded in Quantum Mechanical laws, and not in mathematical or computational complexity. In this theoretical study, we are concerned with Continuous Variable (CV) QKD, where information is encoded through the amplitude and phase of a coherent state of light. In our scenario, we assume that Eve performs a collective attack, where they utilize a beamsplitter to redirect part of each transmitted state to themselves, store it in a quantum memory, and later perform a joint measurement after Alice and Bob communicate classically to announce their outcomes. We show that squeezing, in addition to amplitude and phase modulation, can enhance the security of CV-QKD by enhancing key rates under both imperfect direct and reverse reconciliation (DR/RR) schemes.

Acknowledgments

I would like to express my gratitude to Professor Ioannis Krikidis for hosting me at the IRIDA research center, for supervising this project, and for introducing me to the exciting field of Quantum Optical Communications, where I got to utilize my knowledge in Quantum Optics to investigate an interesting physics and communication problem. I am also grateful towards Dr. Nikos Mitsiou for his insightful feedback. Last but not least, I would like to thank Kyproula Mitsidi for our fruitful collaboration and endless discussions throughout this project.

Contents

1 Introduction **2**
- 1.1 Description of the Protocol . . . 2
  - 1.1.1 Direct and Reverse Reconciliation . . . 2
  - 1.1.2 The role of Eve . . . 2
  - 1.1.3 The role of Squeezing . . . 3
- 1.2 What is the Key Rate? . . . 4
  - 1.2.1 How much information do Alice and Bob know about each other? . . . 4
  - 1.2.2 What is the maximum amount of information Eve can extract? . . . 4

2 Methods **5**
- 2.1 Mutual Information between Alice and Bob . . . 5
- 2.2 Eve's Holevo Bound . . . 5
  - 2.2.1 From Entropy to Covariance Matrices . . . 6
  - 2.2.2 Derivation of Eve's Covariance Matrices . . . 7
- 2.3 Final formula for the Key Rate . . . 9

3 Results **9**
- 3.1 Mutual Information . . . 9
- 3.2 Direct Reconciliation . . . 10
  - 3.2.1 Holevo Bound . . . 10
  - 3.2.2 Key Rate . . . 10
  - 3.2.3 Security benefits from Squeezing . . . 11
- 3.3 Reverse Reconciliation . . . 11
  - 3.3.1 Holevo Bound . . . 11
  - 3.3.2 Key Rate . . . 11
  - 3.3.3 Security benefits from Squeezing . . . 12

4 Conclusions **12**

1 Introduction

1.1 Description of the Protocol

In the CV-QKD protocol under consideration [1], Alice prepares a coherent state with independent amplitude and phase modulation

$$\text{Alice: } \hat{D}(\alpha)\,|0\rangle = \hat{D}\left(\frac{a_x + i a_p}{2}\right)|0\rangle = \left|\frac{a_x + i a_p}{2}\right\rangle = |\alpha\rangle \ , \tag{1}$$

where $a_x,\ a_p$ are drawn from a Normal distribution with variance V_a

$$a_x, a_p \sim \mathcal{N}(0, V_a) \ . \tag{2}$$

Alice's quadratures $\hat{\boldsymbol{r}}_A = (\hat{x}_A, \hat{p}_A)^{\mathsf{T}}$ are given by

$$\hat{\boldsymbol{r}}_A = a_r + \delta\hat{\boldsymbol{r}}_{\text{vac}} \quad \text{(classical modulation + quantum fluctuations)} \ . \tag{3}$$

Next, they transmit this state to Bob through a quantum channel, who performs a homodyne measurement [2] of the $\hat{x}$ or $\hat{p}$ quadrature randomly[1]. After N transmissions, Alice has a vector $\boldsymbol{A}$ of continuous values which represent the amplitude and phase modulations (a_x, a_p) they used, while Bob has a vector $\boldsymbol{B}$ of continuous values which represent their homodyne measurements m_B (where $m = x$ or p)

$$\text{Alice: } \boldsymbol{A} = ((a_x, a_p)_1, \ldots, (a_x, a_p)_N)^{\mathsf{T}} \in \mathbb{R}^{2N} \ , \tag{4}$$

$$\text{Bob: } \boldsymbol{B} = (m_{B_1}, \ldots, m_{B_N})^{\mathsf{T}} \in \mathbb{R}^{N} \ . \tag{5}$$

The vectors $\boldsymbol{A}, \boldsymbol{B}$ are called raw keys. Next, Alice and Bob communicate over a classical channel and discard their values of amplitude or phase modulation and homodyne measurements respectively based on a common rule. This process is called sifting, and turns their raw keys into sifted keys.

1.1.1 Direct and Reverse Reconciliation

Following the sifting process, Alice and Bob perform an error correction algorithm referred to as reconciliation [3], to turn their different sifted keys into one identical shared key. When correction is carried out by taking Alice's sifted key as a reference, the process is known as Direct Reconciliation (DR), while the opposite process is called Reverse Reconciliation (RR). Like every error correction algorithm, DR and RR are not perfect, and thus we introduce the DR and RR efficiencies, defined as $\mathcal{B}_D$ and $\mathcal{B}_R$ respectively. In a realistic QKD scenario, further privacy amplification protocols will be performed to turn the shared key into a secret shared key [4], but we are not concerned with this process in the current study.

1.1.2 The role of Eve

At this point we introduce Eve, an eavesdropper which conducts a collective beam splitter attack (CBSA) to secretly obtain information from Alice's and Bob's communication. In a CBSA scenario, Eve deploys a beamsplitter with transmissivity η in the communication channel and stores the reflected signal in a quantum memory [1]. After the entire transmission process is over, Eve performs a collective (joint) measurement on all the stored states to receive the key. The beamsplitter has the following effect on Bob's and Eve's modes [5]

$$\begin{pmatrix} \hat{a}_B \\ \hat{a}_E \end{pmatrix} = \begin{pmatrix} \sqrt{\eta} & \sqrt{1-\eta} \\ -\sqrt{1-\eta} & \sqrt{\eta} \end{pmatrix} \begin{pmatrix} \hat{a}_A \\ \hat{a}_{\text{vac}} \end{pmatrix} \ , \tag{6}$$

where subscripts $A, B, E,$ vac denote Alice's, Bob's, Eve's and vacuum modes respectively. The quadratures of each participant i, $\hat{\boldsymbol{r}}_i = (\hat{x}_i, \hat{p}_i)^{\mathsf{T}}$ are transformed in the same way. With this information in mind, we can derive the distributions that describe Bob's homodyne measurements

[1] Throughout the report, we assume the following convention for the quadratures: $\hat{x}_\theta = \hat{a}e^{-i\theta} + \hat{a}^\dagger e^{i\theta}$.

$$x_B \sim \mathcal{N}(\sqrt{\eta}a_x,\ \eta V_a + 1)\ , \tag{7}$$
$$p_B \sim \mathcal{N}(\sqrt{\eta}a_p,\ \eta V_a + 1)\ , \tag{8}$$

where we used the fact that $\langle \delta\hat{\boldsymbol{r}}_{\text{vac}} \rangle = 0$ and $\text{Var}(\delta\hat{\boldsymbol{r}}_{\text{vac}}) = 1$. Therefore, Bob perceives Eve's presence as a pure-loss channel with loss parameter $\sqrt{\eta}$ [6].

1.1.3 The role of Squeezing

In this study, we are concerned with whether squeezing prior to amplitude and phase modulation enhances the security of the protocol. At this stage, this intuitively means that Alice and Bob can establish a common key more efficiently with squeezing, after accounting for Eve's presence and reconciliation imperfections. Alice now prepares the following state

$$\text{Alice: } \hat{D}(\alpha)\hat{S}(r)\,|0\rangle = \hat{D}\left(\frac{a_x + i a_p}{2}\right)\hat{S}(r)\,|0\rangle = \left|\frac{a_x + i a_p}{2}, r\right\rangle = |\alpha, r\rangle\ , \tag{9}$$

where the squeezing is applied in the x direction ($r \in \mathbb{R}^+$) without loss of generality. Alice's quadratures are now given by

$$\hat{\boldsymbol{r}}_A = a_r + \delta\hat{\boldsymbol{r}}_{\text{squeezing}} \quad \text{(classical modulation + quantum fluctuations)}\ . \tag{10}$$

Squeezing does not change the mean of the quadrature distribution but affects their variance

$$\text{Var}(\hat{x}_A) = V_a + e^{-2r} = V_a + V\ , \tag{11}$$
$$\text{Var}(\hat{p}_A) = V_a + e^{+2r} = V_a + V^{-1}\ , \tag{12}$$

where we used the fact that $\langle \delta\hat{\boldsymbol{r}}_{\text{squeezing}} \rangle = 0$, $\text{Var}(\delta\hat{\boldsymbol{x}}_{\text{squeezing}}) = V$ and $\text{Var}(\delta\hat{\boldsymbol{p}}_{\text{squeezing}}) = V^{-1}$. In other words, the variance is squeezed in the x quadrature by amount $V \in (0, 1]$ and anti-squeezed in the p quadrature by amount $V^{-1} \in [1, \infty)$. Bob's homodyne measurements are now distributed according to

$$x_B \sim \mathcal{N}(\sqrt{\eta}a_x,\ \eta(V_a + V - 1) + 1)\ , \tag{13}$$
$$p_B \sim \mathcal{N}(\sqrt{\eta}a_p,\ \eta(V_a + V^{-1} - 1) + 1)\ . \tag{14}$$

Evidently, when Alice squeezes the x direction, we estimate that the shared key rate will be improved in the cases where Bob measures the x quadrature. Thus, in the sifting process, they discard all the values of $\boldsymbol{A}, \boldsymbol{B}$ where the latter did not happen. A schematic of the setup [1] is depicted in Fig. 1.

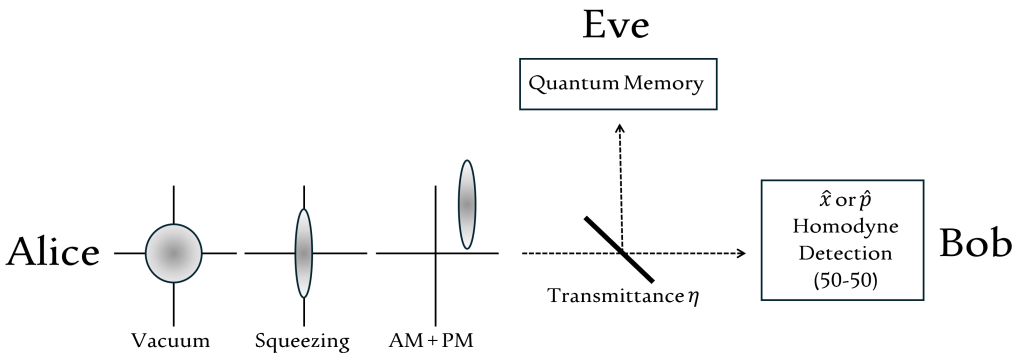


Figure 1: The setup: Alice squeezes the x quadrature, followed by amplitude and phase modulation. They transmit the state to Bob, which is intercepted by Eve's beamsplitter with transmissivity η. The reflected signal is stored in a quantum memory to be measured later. Bob performs an equiprobable homodyne measurement in the x or p direction of their remaining signal.

1.2 What is the Key Rate?

The key rate, which we aim to improve through squeezing, is the amount of secure shared key that Alice and Bob can produce after accounting for Eve and reconciliation imperfections. In order to calculate this quantity, we need the amount of information that Alice and Bob know about each other, minus the maximum amount of information that Eve can extract with their CBSA, all in bits per transmission. The key rate is given by the following formulas for DR and RR

$$\mathcal{K}_D = \mathcal{B}_D \mathcal{I}_{AB} - \chi_{EA} \ , \tag{15}$$
$$\mathcal{K}_R = \mathcal{B}_R \mathcal{I}_{AB} - \chi_{EB} \ . \tag{16}$$

In the following subsections we explore the relevant concepts more deeply.

1.2.1 How much information do Alice and Bob know about each other?

In order to give an answer to this question, we quantify it with the measure of mutual information [7], which is defined below

$$\mathcal{I}_{AB} = H(B) - H(B|A) = H(A) - H(A|B) \ , \tag{17}$$

where

$$H(X) = -\int dx p(x) \log_2 p(x) \tag{18}$$

is the differential Shannon entropy [7], which quantifies the amount of uncertainty in a random variable X, described by probability density function $p(x)$. Intuitively, the mutual information between Alice and Bob is given by the uncertainty in Bob's homodyne measurements, minus the remaining uncertainty in light of Alice's displacements, effectively removing the classical modulation uncertainty V_a. In mathematical terms, for squeezing, displacements and homodyne measurements in the x direction, it is given by

$$\mathcal{I}_{AB} = -\int_{-\infty}^{+\infty} dx_B p(x_B) \log_2 p(x_B) + \int_{-\infty}^{+\infty} da_x p(a_x) \left[\int_{-\infty}^{+\infty} dx_B p(x_B|a_x) \log_2 p(x_B|a_x) \right] \ , \tag{19}$$

where

$$p(a_x) = \mathcal{N}(0,\ \sigma_A^2 = V_a) \ , \tag{20}$$
$$p(x_b) = \mathcal{N}(0,\ \sigma_B^2 = \eta(V_a + V - 1) + 1) \ , \tag{21}$$
$$p(x_b|a_x) = \mathcal{N}(\sqrt{\eta} a_x,\ \sigma_B'^2 = \sigma_B^2 - \eta \sigma_A^2) \ . \tag{22}$$

1.2.2 What is the maximum amount of information Eve can extract?

The maximum information that Eve can extract per transmission is quantified by the Holevo Bound [8], and is given by the following expressions for DR and RR

$$\chi_{EA} = S(\rho_E) - S(\rho_{E|a_x}) \quad \text{(Direct Reconciliation)} \ , \tag{23}$$
$$\chi_{EB} = S(\rho_E) - S(\rho_{E|x_B}) \quad \text{(Reverse Reconciliation)} \ , \tag{24}$$

where ρ_E is Eve's density matrix, and $\rho_{E|a_x}$ and $\rho_{E|x_B}$ is Eve's density matrix conditioned on the outcomes of Alice and Bob respectively. $S(\rho)$ denotes the Von-Neumann entropy [9] of a quantum state with density matrix ρ, given by

$$S(\rho) = -\operatorname{Tr}(\rho \log_2 \rho) \ . \tag{25}$$

The Von-Neumann entropy quantifies the quantum uncertainty or mixedness associated with ρ; for pure states, $S(\rho) = 0$. Essentially, the Holevo bound is the uncertainty associated with Eve's quantum state, minus the remaining uncertainty of their state in light of Alice's or Bob's outcomes.

2 Methods

In this section we explain the mathematical proofs that lead to the final formula for the key rate, which is the main result of this project.

2.1 Mutual Information between Alice and Bob

In order to derive the formula for the Mutual Information between Alice and Bob, we need to expand Eq. 19. The first integral, I_1 is

$$I_1 = -\int_{-\infty}^{+\infty} dx_B C_B G(x_B) \log_2[C_B G(x_B)] \ , \tag{26}$$

where $C_B = (2\pi\sigma_B^2)^{-1/2}$ is the normalization factor of the normal distribution $p(x_B)$, and $G(x_B) = p(x_B)/C_B = e^{-x_B^2/2\sigma_B^2}$. Proceeding with the integral,

$$\begin{aligned} I_1 &= -C_B \left(\log_2 C_B \int_{-\infty}^{+\infty} dx_B G(x_B) + \int_{-\infty}^{+\infty} dx_B G(x_B) \log_2 G(x_B) \right) \\ &= -\log_2 C_B + \frac{C_B \log_2 e}{2\sigma_B^2} \int_{-\infty}^{+\infty} dx_B G(x_B) x_B^2 \\ &= -\log_2 C_B + \frac{\log_2 e}{2} \\ &= \frac{1}{2} \log_2 \left[2\pi e \sigma_B^2 \right] \ . \end{aligned} \tag{27}$$

The second integral, I_2, is

$$I_2 = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} da_x dx_B C_{a_x} C_{B'} G(a_x) G(x_B|a_x) \log_2 \left[C_{B'} G(x_|a_x) \right] \ , \tag{28}$$

Where $C_{a_x} = (2\pi\sigma_A^2)^{-1/2}$ is the normalization factor of the normal distribution $p(a_x)$, and $G(a_x) = p(a_x)/C_{a_x} = e^{-a_x^2/2\sigma_A^2}$. Similarly, $C_{B'} = (2\pi\sigma_{B'}^2)^{-1/2}$ is the normalization factor of the normal distribution $p(x_B|a_x)$, and $G(x_B|a_x) = p(x_B|a_x)/C_{B'} = e^{-(x_B - \sqrt{\eta} a_x)^2/2\sigma_{B'}^2}$. Solving the integral with the same techniques as I_1, we yield

$$I_2 = -\frac{1}{2} \log_2 \left[2\pi e \sigma_{B'}^2 \right] \ . \tag{29}$$

Finally, the mutual information between Alice and Bob is given by

$$\begin{aligned} \mathcal{I}_{AB} &= I_1 + I_2 \\ &= \frac{1}{2} \log_2 \left[\frac{\sigma_B^2}{\sigma_{B'}^2} \right] \\ &= \frac{1}{2} \log_2 \left[\frac{\eta(V_a + V - 1) + 1}{\eta(V-1) + 1} \right] \\ &= \frac{1}{2} \log_2 \left[1 + \frac{\eta V_a}{\eta(V-1) + 1} \right] \ . \end{aligned} \tag{30}$$

2.2 Eve's Holevo Bound

Eve's Holevo bound involves several diffuclt and cumbersome calculations with density matrices; for this reason, in the following section we lay down some mathematics to prove that we can use covariance matrices (CMs) to get the result we are after, instead.

2.2.1 From Entropy to Covariance Matrices

When a state has diagonal density matrix, the argument of the trace in Eq. 25 becomes

$$\begin{aligned}\rho \log_2 \rho &= \left(\sum_n \lambda_n |n\rangle\langle n|\right)\left(\sum_m \log_2 \lambda_m |m\rangle\langle m|\right) \\ &= \sum_n \sum_m \lambda_n \log_2 \lambda_m |n\rangle \langle n|m\rangle \langle m| \\ &= \sum_n \lambda_n \log_2 \lambda_n |n\rangle\langle n| \ . \end{aligned} \tag{31}$$

Then, the entropy is given by

$$S(\rho) = -\operatorname{Tr}(\rho \log_2 \rho) = -\sum_n \sum_k \lambda_n \log_2 \lambda_n \langle k|n\rangle \langle n|k\rangle = -\sum_n \lambda_n \log_2 \lambda_n \ . \tag{32}$$

Such states with a diagonal density matrix and known eigenvalues are the thermal states

$$\rho_{\text{th}} = \sum_n \underbrace{\frac{\bar{n}_{\text{th}}^n}{(\bar{n}_{\text{th}}+1)^{n+1}}}_{\text{eigenvalues } \lambda_n^{\text{th}}} |n\rangle\langle n| \ , \tag{33}$$

where $\bar{n}$ is the average number of photons $\langle \hat{n} \rangle_{\text{th}}$. Throughout the study, we are concerned with single-mode Gaussian states (SMGS), which we can always unitarily transform into a thermal state [10], and extract the corresponding eigenvalues[2]. Then, we can proceed with the calculation of the entropy straightforwardly. To prove this claim, an arbitrary SMGS may be expressed as

$$\rho_{\text{SMGS}}(\omega, \zeta, \bar{n}_{\text{th}}) = \hat{D}(\omega)\hat{S}(\zeta)\rho_{\text{th}}(\bar{n}_{\text{th}})\hat{S}^\dagger(\zeta)\hat{D}^\dagger(\omega) \ , \tag{34}$$

where $\omega, \ \zeta \in \mathbb{C}$ are the displacement and squeezing parameters respectively, and $\bar{n}_{\text{th}} \in \mathbb{R}_0^+$ is the average number of photons of the thermal state. For example, $|\omega\rangle\langle\omega| = \rho_{\text{SMGS}}(\omega, 0, 0)$ and $|\zeta\rangle\langle\zeta| = \rho_{\text{SMGS}}(0, \zeta, 0)$. We can transform the arbitrary SMGS into a thermal state by undoing the displacement and squeezing

$$\rho'_{\text{SMGS}} \to \hat{U}\rho_{\text{SMGS}}\hat{U}^\dagger = \rho_{\text{th}}, \ \text{where } \hat{U} = \hat{S}^\dagger(\zeta)\hat{D}^\dagger(\omega) \ . \tag{35}$$

Upon inspection of the above alongside Eq. 32, we derive the following, very important result

$$S(\rho_{\text{SMGS}}) = S(\rho_{\text{th}}) = -\sum_n \lambda_n^{\text{th}} \log_2 \lambda_n^{\text{th}} = \mathcal{G}(\bar{n}_{\text{th}}) \ , \tag{36}$$

where

$$\mathcal{G}(x) = (x+1)\log_2(x+1) - x \log_2 x \ . \tag{37}$$

This means that if we have an arbitrary SMGS, ρ_{SMGS}, we can derive the entropy by finding the average number of photons $\bar{n}_{\text{th}}$ of the equivalent thermal state defined by Eq. 35. To find the number of photons $\bar{n}_{\text{th}}$ of the equivalent thermal state for our arbitrary SMGS, we proceed by computing its CM

$$\boldsymbol{V}_{\text{SMGS}} = \begin{pmatrix} \text{Var}(\hat{x}) & \text{Cov}(\hat{x}, \hat{p}) \\ \text{Cov}(\hat{x}, \hat{p}) & \text{Var}(\hat{p}) \end{pmatrix} = \begin{pmatrix} \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2 & \frac{1}{2}\langle \hat{x}\hat{p} + \hat{p}\hat{x} \rangle - \langle \hat{x} \rangle \langle \hat{p} \rangle \\ \frac{1}{2}\langle \hat{x}\hat{p} + \hat{p}\hat{x} \rangle - \langle \hat{x} \rangle \langle \hat{p} \rangle & \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2 \end{pmatrix} \ . \tag{38}$$

Through Williamson's theorem [10], we know that the CM of an arbitrary SMGS may be decomposed in the following way

[2] The eigenvalues of a density matrix remain the same under a unitary transformation.

$$\boldsymbol{V}_{\text{SMGS}} = \boldsymbol{S} \begin{pmatrix} \nu & 0 \\ 0 & \nu \end{pmatrix} \boldsymbol{S}^{\mathsf{T}} \ , \tag{39}$$

where $\boldsymbol{S}$ is a symplectic transformation. Next, we take the determinant on both sides

$$\det \boldsymbol{V}_{\text{SMGS}} = \cancelto{1}{(\det \boldsymbol{S})}(\det \nu \boldsymbol{I}) \cancelto{1}{(\det \boldsymbol{S}^{\mathsf{T}})} = \nu^2 \Rightarrow \nu = \sqrt{\det \boldsymbol{V}_{\text{SMGS}}} \ . \tag{40}$$

Interestingly, for the appropriate value of ν, $\nu \boldsymbol{I}$ is uniquely defined as the CM of a thermal state

$$\boldsymbol{V}_{\text{th}} = (2\bar{n}_{\text{th}} + 1)\boldsymbol{I} \Rightarrow \nu = 2\bar{n}_{\text{th}} + 1 \ . \tag{41}$$

Thus, upon inspection of Eq. 40 alongside the above result, an arbitrary SMGS is equivalent to a thermal state with the following number of photons

$$\bar{n}_{\text{th}} = \frac{\sqrt{\det \boldsymbol{V}_{\text{SMGS}}} - 1}{2} \ . \tag{42}$$

Finally, by combining this result with Eq. 36, the entropy of an arbitrary SMGS is given by

$$S(\rho_{\text{SMGS}}) = \mathcal{G}\left(\frac{\sqrt{\det \boldsymbol{V}_{\text{SMGS}}} - 1}{2} \right) \ . \tag{43}$$

Therefore, the problem of calculating the entropy of Eve's states in different scenarios is effectively reduced to the calculation of the corresponding CMs.

2.2.2 Derivation of Eve's Covariance Matrices

We have three different situations in which we will calculate CMs (in the form of Eq. 38): for Eve's state in the complete absence of knowledge regarding Alice's and Bob's outcomes, and for Eve's state conditioned on the outcomes of Alice and Bob separately. To calculate said CMs, we need Eve's quadratures $\hat{\boldsymbol{r}}_E = (\hat{x}_E, \hat{p}_E)^{\mathsf{T}}$, which are given with respect to the input modes according to the beamsplitter transformation (Eq. 6)

$$\hat{\boldsymbol{r}}_E = -\sqrt{1-\eta}\hat{\boldsymbol{r}}_A + \sqrt{\eta}\hat{\boldsymbol{r}}_{\text{vac}} \ , \tag{44}$$

where E, A, vac denote Eve's, Alice's, and the vacuum mode respectively.

CM Unconditioned on any outcome The variances of Eve's quadratures are

$$\text{Var}(\hat{x}_E) = (1-\eta)\,\text{Var}(\hat{x}_A) + \eta\,\text{Var}(\hat{x}_{\text{vac}}) = (1-\eta)(V + V_a) + \eta \ , \tag{45}$$
$$\text{Var}(\hat{p}_E) = (1-\eta)\,\text{Var}(\hat{p}_A) + \eta\,\text{Var}(\hat{p}_{\text{vac}}) = (1-\eta)(V^{-1} + V_a) + \eta \ . \tag{46}$$

We proceed with the covariance terms by expanding the formula $\text{Cov}(\hat{x}_E, \hat{p}_E) = \frac{1}{2}\langle \hat{x}_E\hat{p}_E + \hat{p}_E\hat{x}_E \rangle - \langle \hat{x}_E \rangle \langle \hat{p}_E \rangle$ in terms of the Alice and vacuum modes

$$\text{Cov}(\hat{x}_E, \hat{p}_E) = \frac{1}{2}\left\langle (1-\eta)\{\hat{x}_A, \hat{p}_A\} + \sqrt{\eta(1-\eta)}\left(\{\hat{x}_A, \hat{p}_{\text{vac}}\} - \{\hat{x}_{\text{vac}}, \hat{p}_A\}\right) + \eta\{\hat{x}_{\text{vac}}, \hat{p}_{\text{vac}}\} \right\rangle - (1-\eta)a_x a_p \ . \tag{47}$$

Note that $\{\cdot\}$ denotes the anti-commutator, and all expectation values are taken with respect to the input modes $|\alpha, r\rangle_A \otimes |0\rangle_{\text{vac}}$. Then, by expanding the quadrature operators, we notice that all terms which involve the vacuum mode contain operations of the type $\left\langle \hat{a}_{\text{vac}} \text{ or } \hat{a}^{\dagger}_{\text{vac}} \text{ or } \hat{a}^2_{\text{vac}} \text{ or } (\hat{a}^{\dagger}_{\text{vac}})^2 \text{ or } \hat{n}_{\text{vac}} \right\rangle_{\text{vac}} = 0$. Therefore, we are left with

$$\text{Cov}(\hat{x}_E, \hat{p}_E) = \frac{(1-\eta)}{2}\left(\langle\{\hat{x}_A, \hat{p}_A\}\rangle - 2a_x a_p\right) = i(1-\eta)\left(\alpha^2 - (\alpha^*)^2 - \text{Re}(\alpha)\,\text{Im}(\alpha)\right) = 0 \ . \tag{48}$$

Hence, Eve's CM unconditioned on any outcome is

$$\boldsymbol{V}_E = \begin{pmatrix} (1-\eta)(V + V_a) + \eta & 0 \\ 0 & (1-\eta)(V^{-1} + V_a) + \eta \end{pmatrix} \ . \tag{49}$$

CM Conditioned on Alice's outcome We now investigate how Eve's CM changes in light of Alice's outcome. We remind that Alice keeps their amplitude or phase modulation, a_x or a_p respectively, depending on Bob's homodyne measurement. In the proceeding proof, we assume that Alice keeps a_x; the proof in the opposite case yields identical results. Once Eve knows Alice's a_x, there is no longer uncertainty due to Alice's classical amplitude modulation, and their variance of $\hat{x}_E$ is now given by [11]

$$\mathrm{Var}(\hat{x}_E|a_x) = \mathrm{Var}(\hat{x}_E) - \frac{\mathrm{Cov}(\hat{x}_E, a_x)^2}{\mathrm{Var}(a_x)} \ . \tag{50}$$

We proceed by calculating the subtracted term

$$\begin{aligned}
\frac{\mathrm{Cov}(\hat{x}_E, a_x)^2}{\mathrm{Var}(a_x)} &= \frac{1}{V_a}\left(\frac{\langle \hat{x}_E a_x + a_x \hat{x}_E\rangle}{2} - \langle \hat{x}_E\rangle \cancelto{0}{\langle a_x\rangle}\right)^2 \\
&= \frac{1}{V_a}\left(\left\langle \sqrt{\eta}\cancelto{0}{\hat{x}_{\mathrm{vac}} a_x} - \sqrt{1-\eta}\hat{x}_A a_x \right\rangle\right)^2 \\
&= \frac{1-\eta}{V_a}\left(\langle a_x^2\rangle\right)^2 = \frac{1-\eta}{V_a}\left(\mathrm{Var}(a_x) - \cancelto{0}{\langle a_x\rangle}\right)^2 = \frac{1-\eta}{V_a}V_a^2 \\
&= (1-\eta)V_a \ .
\end{aligned} \tag{51}$$

Since $\hat{\rho}_E$ is independent of $\hat{x}_A$, the corresponding component is not altered, and the CM in this case is

$$\boldsymbol{V}_{E|a_x} = \begin{pmatrix} (1-\eta)V + \eta & 0 \\ 0 & (1-\eta)(V^{-1} + V_a) + \eta \end{pmatrix} \ . \tag{52}$$

Conditioned on Bob's outcome When Bob performs a homodyne measurement on the x quadrature and yields the result x_B, Eve's variance of $\hat{x}_E$ changes in the following way

$$\mathrm{Var}(\hat{x}_E|x_B) = \mathrm{Var}(\hat{x}_E) - \frac{\mathrm{Cov}(\hat{x}_E, x_B)^2}{\mathrm{Var}(x_B)} \ . \tag{53}$$

The variance of x_B can be derived from the beamsplitter transformations (Eq. 6)

$$\hat{x}_B = \sqrt{\eta}\hat{x}_A + \sqrt{1-\eta}\hat{x}_{\mathrm{vac}} \tag{54}$$

$$\Rightarrow \mathrm{Var}(\hat{x}_B) = \eta(V + V_a) + (1-\eta) \ . \tag{55}$$

The covariance term for an ideal homodyne measurement is $\mathrm{Cov}(\hat{x}_E, x_B) = \mathrm{Cov}(\hat{x}_E, \hat{x}_B)$. Upon expanding, it becomes

$$\begin{aligned}
\mathrm{Cov}(\hat{x}_E, \hat{x}_B) &= \frac{1}{2}\langle \hat{x}_E\hat{x}_B - \hat{x}_B\hat{x}_E\rangle - \langle \hat{x}_E\rangle\langle \hat{x}_B\rangle \\
&= \sqrt{\eta(1-\eta)}(1 - \langle x_A^2\rangle) = \sqrt{\eta(1-\eta)}(1 - \mathrm{Var}(x_A)) \\
&= \sqrt{\eta(1-\eta)}(1 - V - V_a) \ .
\end{aligned} \tag{56}$$

We again used the fact that operations involving the vacuum mode are zero. Eve's variance in $\hat{x}_E$ is then

$$\mathrm{Var}(\hat{x}_E|x_B) = (1-\eta)(V + V_a) + \eta - \frac{\eta(1-\eta)(1 - V - V_a)^2}{\eta(V + V_a) + (1-\eta)} \ . \tag{57}$$

Since $\hat{\rho}_E$ is independent of x_B, the corresponding component is not altered, and the CM in this case is

$$\boldsymbol{V}_{E|x_B} = \begin{pmatrix} (1-\eta)(V + V_a) + \eta - \frac{\eta(1-\eta)(1-V-V_a)^2}{\eta(V+V_a)+(1-\eta)} & 0 \\ 0 & (1-\eta)(V^{-1} + V_a) + \eta \end{pmatrix} \ . \tag{58}$$

2.3 Final formula for the Key Rate

We now have both ingredients to construct the final formula for the key rate, namely Alice's and Bob's mutual information $\mathcal{I}_{AB}$ and Eve's Holevo bounds $\chi_{EA/EB}$. Below, we present formulas for both DR and RR, which are a combination of Eqs. 30, 43

Direct Reconciliation:

$$\mathcal{K}_D = \frac{\mathcal{B}_D}{2}\log_2\left[1+\frac{\eta V_a}{\eta(V-1)+1}\right] - \left(\frac{\sqrt{\det \boldsymbol{V}_E}+1}{2}\right)\log_2\left(\frac{\sqrt{\det \boldsymbol{V}_E}+1}{2}\right) + \left(\frac{\sqrt{\det \boldsymbol{V}_E}-1}{2}\right)\log_2\left(\frac{\sqrt{\det \boldsymbol{V}_E}-1}{2}\right)$$
$$+ \left(\frac{\sqrt{\det \boldsymbol{V}_{E|a_x}}+1}{2}\right)\log_2\left(\frac{\sqrt{\det \boldsymbol{V}_{E|a_x}}+1}{2}\right) - \left(\frac{\sqrt{\det \boldsymbol{V}_{E|a_x}}-1}{2}\right)\log_2\left(\frac{\sqrt{\det \boldsymbol{V}_{E|a_x}}-1}{2}\right) \tag{59}$$

Reverse Reconciliation:

$$\mathcal{K}_R = \frac{\mathcal{B}_R}{2}\log_2\left[1+\frac{\eta V_a}{\eta(V-1)+1}\right] - \left(\frac{\sqrt{\det \boldsymbol{V}_E}+1}{2}\right)\log_2\left(\frac{\sqrt{\det \boldsymbol{V}_E}+1}{2}\right) + \left(\frac{\sqrt{\det \boldsymbol{V}_E}-1}{2}\right)\log_2\left(\frac{\sqrt{\det \boldsymbol{V}_E}-1}{2}\right)$$
$$+ \left(\frac{\sqrt{\det \boldsymbol{V}_{E|x_B}}+1}{2}\right)\log_2\left(\frac{\sqrt{\det \boldsymbol{V}_{E|x_B}}+1}{2}\right) - \left(\frac{\sqrt{\det \boldsymbol{V}_{E|x_B}}-1}{2}\right)\log_2\left(\frac{\sqrt{\det \boldsymbol{V}_{E|x_B}}-1}{2}\right) \tag{60}$$

3 Results

3.1 Mutual Information

We firstly present in Fig. 2 the result for the mutual information between Alice and Bob for varying transmissivities $\eta = \{0.0, 0.1, \ldots, 1.0\}$, $\mathcal{I}_{AB}(V_a, V|\eta)$, which is invariant of the reconciliation scheme.

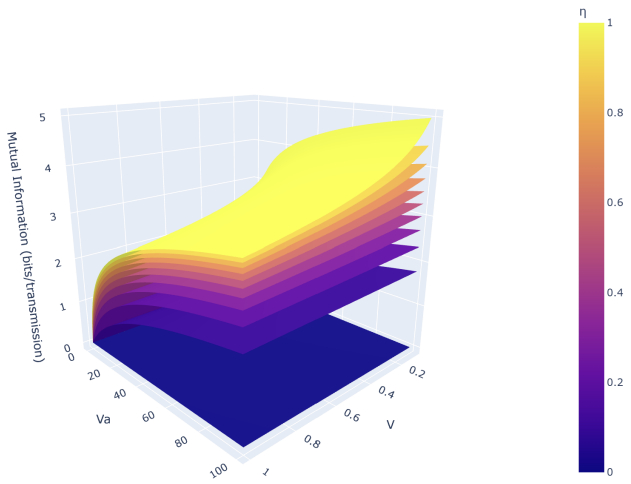


Figure 2: Mutual information $\mathcal{I}_{AB}(V_a, V|\eta)$ as a function of classical modulation V_a and squeezing variance V for varying transmissivities $\eta = \{0.0, 0.1, \ldots, 1.0\}$.

As expected, higher transmissivity η corresponds to more information being sent to Bob, and thus higher mutual information or correlation between themselves and Alice. An interesting observation is that more squeezing ($r \to \infty$, $V \to 0$) continuously increases the mutual information, but only for high values of transmissivity η. Squeezing decreases the variance in Bob's homodyne measurement distribution $p(x_B)$, and thus leads to better correlations with Alice, only if they get an adequately strong signal; low transmissivity η means a weak signal for Bob, and squeezing does not provide any additional improvements.

3.2 Direct Reconciliation

3.2.1 Holevo Bound

The Holevo bound for Eve in the DR scheme is depicted in Fig. 3.

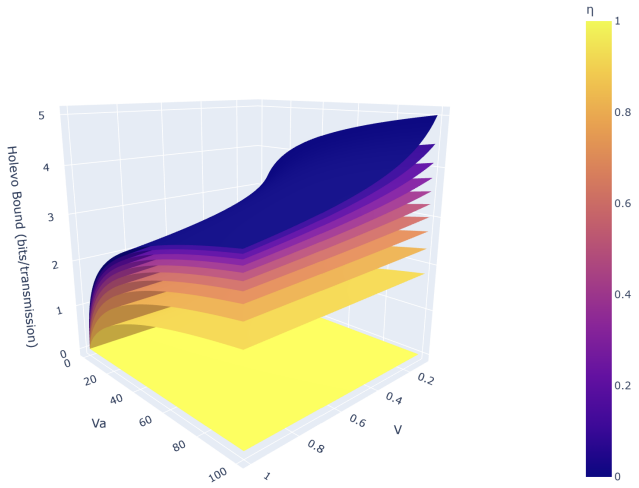


Figure 3: Eve's Holevo bound in the DR scheme, $\chi_{EA}(V_a, V|\eta)$ for varying transmissivities $\eta = \{0.0, 0.1, \ldots, 1.0\}$.

Lower transmissivity η corresponds to more information being kept by Eve, and thus higher amount of information they can extract from Alice. Again, an interesting observation is that more squeezing ($r \to \infty$, $V \to 0$) continuously increases the Holevo bound, but only for low values of transmissivity η; this means that the eavesdropper actually benefits from squeezing in the low transmissivity regime. Squeezing decreases the variance in Eve's joint homodyne measurement distribution, leading to stronger correlations with Alice. In the high transmissivity regime, the Holevo bound is independent from squeezing.

3.2.2 Key Rate

The key rate $\mathcal{K}_D(V_a, V|\eta, \mathcal{B}_D)$ is depicted in Fig. 4, as a function of classical modulation V_a and squeezing variance V for constant transmissivity $\eta = 0.6$ and varying DR efficiencies $\mathcal{B}_D = \{0.80, 0.81, \ldots, 0.95\}$. We observe that for each surface, there is a specific curve of the classical modulation $V_a(V)$ which maximizes key rate, while increased squeezing ($r \to \infty, V \to 0$) continuously improves the key rate for any value of the classical modulation V_a.

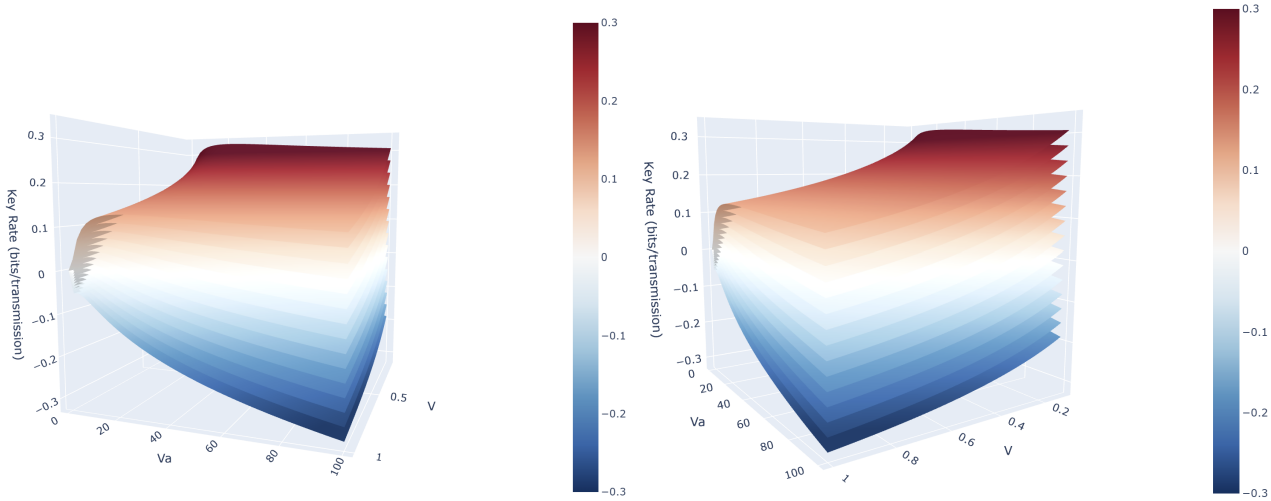


Figure 4: Key rate $\mathcal{K}_D(V_a, V|\eta, \mathcal{B}_D)$ as a function of classical modulation V_a and squeezing variance V for constant transmissivity $\eta = 0.6$ and varying DR efficiencies $\mathcal{B}_D = \{0.80, 0.81, \ldots, 0.95\}$. Blue and red colors correspond to negative and positive key rates respectively.

.

3.2.3 Security benefits from Squeezing

We further explore whether squeezing can produce positive key rates, necessary for secure communication, across the entire spectrum of η and $\mathcal{B}_D$ (both $\in [0, 1]$). The results are depicted in Fig 5: On the left, we see a security map classifying the $\eta - \mathcal{B}_D$ regions based on the role of squeezing. In the green region, coherent states ($r = 0, V = 1$) can produce positive key rates, but squeezing ($r > 0, V < 1$) is useful for further increasing them. In the yellow region, squeezing is necessary for positive key rates, as coherent states are unable to produce them. Lastly, in the red region, neither coherent nor displaced squeezed states can produce positive key rates. The black surface is the zero plane. On the right, there are three color coded key rate surfaces for $(\eta, \mathcal{B}_D)$ points from the corresponding regions on the left. We observe that the green region starts at approximately $\eta \approx 0.5$ and continuously expands for lower $\mathcal{B}_D$ with as $\eta \to 1$. Squeezing extends the positive key rate (yellow) region into lower reconciliation efficiencies, providing a significant improvement. Secure communication is impossible for any value of $\mathcal{B}_D$ for $\eta < 0.5$.

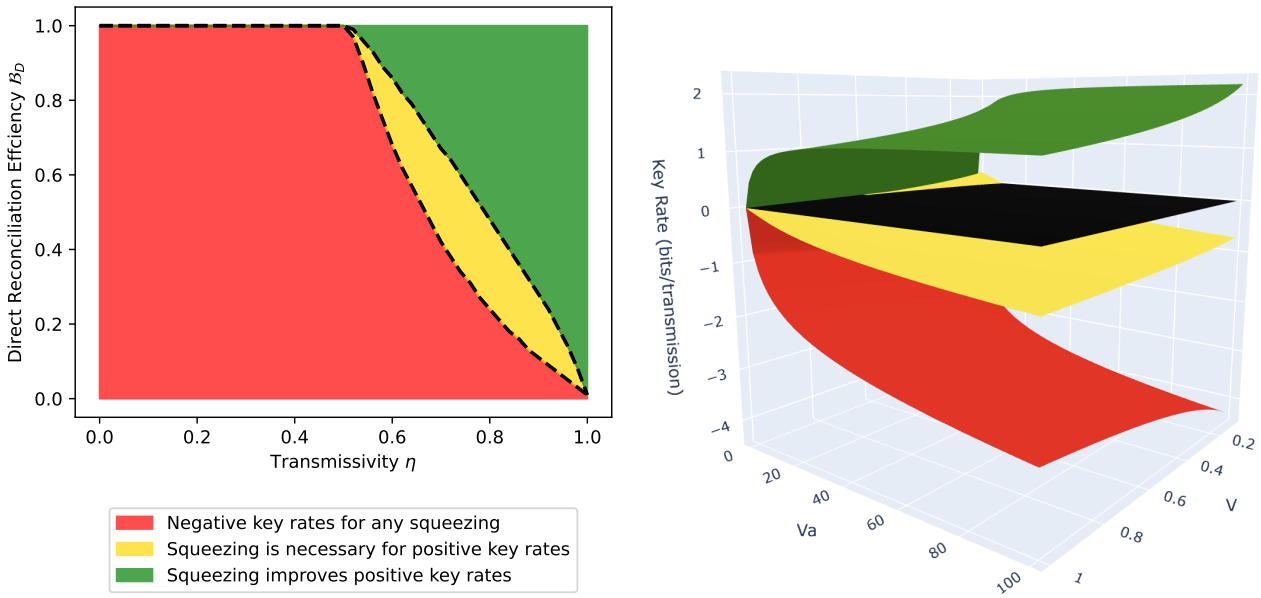


Figure 5: **(Left)** Security map: squeezing cannot help produce positive key rates (red), squeezing is necessary for positive key rates (yellow), squeezing improves positive key rates (green), as a function of $\mathcal{B}_D$ and η. **(Right)** Key rates $\mathcal{K}_D(V_a, V|\eta, \mathcal{B}_D)$ for $(\eta, \mathcal{B}_D)$ points taken from the corresponding colored regions on the left plot. The black plane represents the zero surface.

3.3 Reverse Reconciliation

3.3.1 Holevo Bound

The Holevo bound for Eve in the RR scheme is depicted in Fig. 6. We observe that in this case the Holevo bound starts from zero in the low transmissivity regime, peaks at $\eta \approx 0.5$, and then approaches zero again in the high transmissivity regime. For $\eta \approx 0.0$, no information gets transmitted to Bob, and thus it is impossible for Eve to establish correlations and extract information from them. For $\eta \approx 1.0$, Eve gets no signal from their beamsplitter whereas Bob has all of it, disallowing Eve from establishing correlations with Bob. For $\eta \approx 0.5$, the information is split evenly between Bob and Eve, and thus Eve can extract the maximum amount of information from Bob. In contrast with DR, now the Holevo bound is independent from squeezing. The latter only helps Bob establish better correlations with Alice in both reconciliation schemes by narrowing the variance of their homodyne measurement distribution. The same effect also helps Eve in the DR scheme.

3.3.2 Key Rate

The key rate $\mathcal{K}_R(V_a, V|\eta, \mathcal{B}_R)$ is depicted in fig. 7 as a function of classical modulation V_a and squeezing variance V for constant transmissivity $\eta = 0.6$ and varying RR efficiencies $\mathcal{B}_R = \{0.35, 0.40, \ldots, 0.95\}$. Similarly with the DR case, we observe that for each surface, there is a specific curve of the classical modulation $V_a(V)$ which maximizes key rate, while increased squeezing ($r \to \infty, V \to 0$) continuously improves the key rate for any value of the classical modulation V_a.

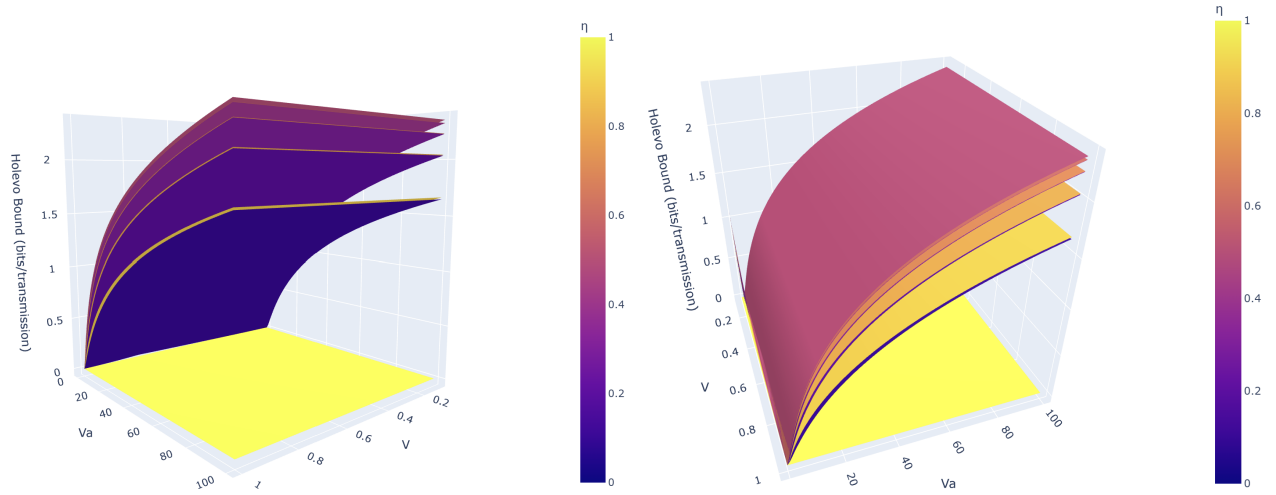


Figure 6: Eve's Holevo bound in the RR scheme, $\chi_{EB}(V_a, V|\eta)$ for varying transmissivities $\eta = \{0.0, 0.1, \ldots, 1.0\}$.

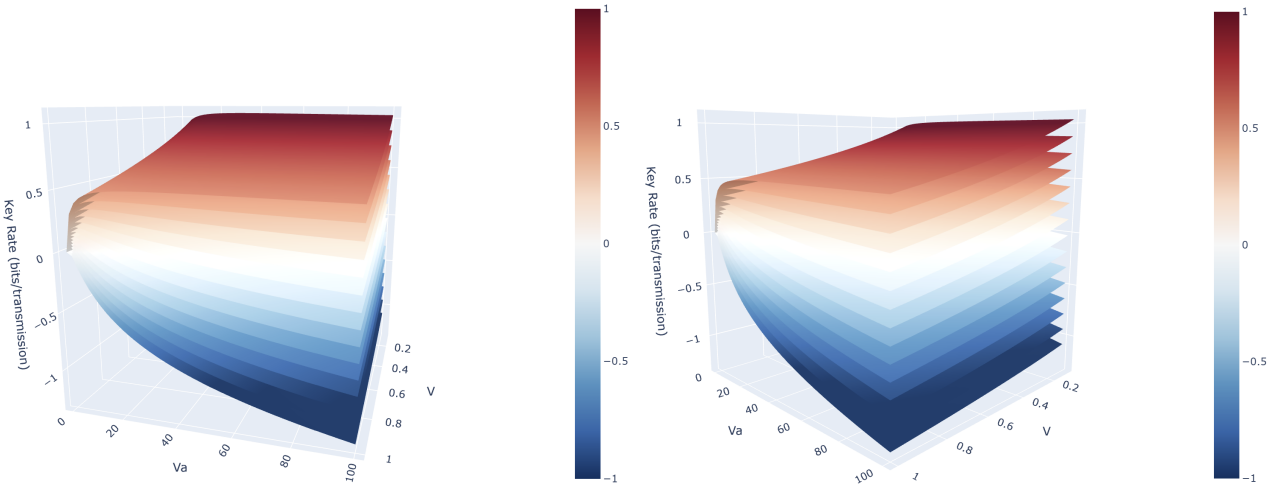


Figure 7: Key rate $\mathcal{K}_R(V_a, V|\eta, \mathcal{B}_R)$ as a function of classical modulation V_a and squeezing variance V for constant transmissivity $\eta = 0.6$ and varying RR efficiencies $\mathcal{B}_R = \{0.35, 0.40, \ldots, 0.95\}$. Blue and red colors correspond to negative and positive key rates respectively.

.

3.3.3 Security benefits from Squeezing

The results for the security benefits from squeezing are depicted for the RR scheme in fig 8. We observe that the green region exists for almost all values of η and continuously expands for lower RR efficiencies $\mathcal{B}_R$ with increasing transmissivity. There can be secure communication with both coherent and displaced squeezed states for transmissivity and RR efficiency as low as $\eta \approx 0.1$ and $\mathcal{B}_R \approx 0.5$ respectively. Squeezing dramatically extends the positive key rate (yellow) region into lower reconciliation efficiencies, effectively allowing secure communication for almost any value of transmissivity η and RR efficiency $\mathcal{B}_R$. Secure communication is impossible only for very low values of the RR efficiency ($\mathcal{B}_R < 10^{-3}, \forall \eta \in [0, 1]$). Again, the black surface is the zero plane, and on the right there are three color coded key rate surfaces for $(\eta, \mathcal{B}_R)$ points from the corresponding regions on the left. The reason for this huge benefit from squeezing with respect to the DR scheme is that now the Holevo bound is independent of squeezing (fig. 6) while the mutual information grows with larger squeezing (fig. 2). Additionally, as stated in sec. 3.3.1., low (high) transmissivity disallows Eve from establishing strong correlations with Bob because Bob does not have enough information (Bob has most of the information), regardless of squeezing.

4 Conclusions

In conclusion, we investigated whether squeezing can provide enhanced key rates and thus improved security when utilized in a CV-QKD protocol with coherent states under collective attack modelled as a beamsplitter with trans-

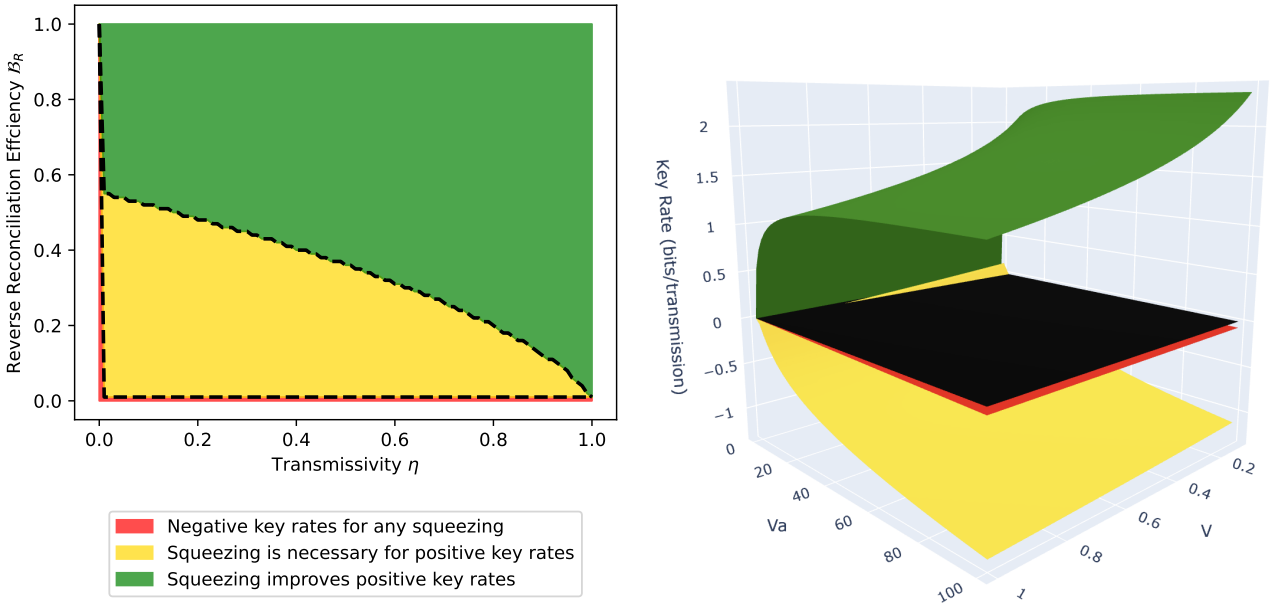


Figure 8: **(Left)** Security map: squeezing cannot help produce positive key rates (red), squeezing is necessary for positive key rates (yellow), squeezing improves positive key rates (green), as a function of RR efficiency $\mathcal{B}_R$ and transmissivity η. **(Right)** Key rates $\mathcal{K}_R(V_a, V|\eta, \mathcal{B}_R)$ for values of transmissivity η and RR efficiency $\mathcal{B}_R$ taken from the corresponding colored region on the left plot. The black plane represents the zero surface.

missivity η and imperfect reconciliation efficiency. More specifically, the transmitter Alice squeezed a vacuum mode, and then performed independent gaussian amplitude and phase modulation, i.e. complex displacement, with a certain variance. As the quantum state traveled to the receiver Bob, it traversed a beamsplitter placed by an eavesdropper Eve, who stored all their stolen signals in a quantum memory to be collectively measured after the completion of the protocol. Upon reception of the remaining signal, Bob performed a homodyne measurement of the amplitude or phase quadrature with equal probability. Afterwards, Alice and Bob communicated over a classical channel, and discarded the amplitude or phase displacements and homodyne measurements respectively, in the cases when Bob did not measure the squeezed quadrature. The key rate was derived by computing the mutual information between Alice and Bob and subtracting the maximum information possibly extracted by Eve, the Holevo bound, after accounting for inefficiencies in the direct or reverse reconciliation process.

We observed that the mutual information information between Alice and Bob grew continuously with squeezing in the large transmissivity regime, as Bob had an increasingly narrower homodyne measurement distribution, approaching Alice's original modulation more accurately. In the low transmissivity regime, squeezing did not provide any measurable benefit to Bob's weak signal. In the DR scheme, the Holevo bound continuously grew with decreasing transmissivity as Eve received a stronger signal, and was further improved by squeezing in the low transmissivity regime for the same reason as before, but now applied to Alice and Eve. For high transmissivities, Alice's signal was so weak that squeezing did not provide any measurable benefits. In the RR scheme, the Holevo bound grew with transmissivity until $\eta \approx 0.5$ and dropped again as transmissivity increased. In the edge scenarios, either Bob or Eve did not hold a strong enough signal to establish any significant correlations between them, and the situation was most balanced for an evenly split signal at $\eta \approx 0.5$. Most importantly, in the RR scheme, the Holevo bound was independent of squeezing. In both reconciliation schemes, the key rate was continuously improved with squeezing, while there was an optimal amount of classical modulation variance for which maximized the key rate.

By exploring the transmissivity-reconciliation efficiency ($\eta - \mathcal{B}$) map for both reconciliation schemes, we showed that positive key rates, and thus secure communication, can only be produced for certain regions of the map. In both cases, these security regions were substantially extended by the utilization of squeezing, albeit the provided benefit was much larger for the RR scheme due to the aforementioned reasons.

References

[1] Vladyslav C. Usenko and Radim Filip. "Squeezed-state quantum key distribution upon imperfect reconciliation". In: *New Journal of Physics* 13.11 (2011), p. 113007. DOI: 10.1088/1367-2630/13/11/113007.

[2] Horace P. Yuen and Vincent W. S. Chan. "Noise in homodyne and heterodyne detection". In: *Optics Letters* 8.3 (1983), pp. 177–179. DOI: 10.1364/OL.8.000177.

[3] Gilles Van Assche, Jean Cardinal, and Nicolas J. Cerf. "Reconciliation of a Quantum-Distributed Gaussian Key". In: *IEEE Transactions on Information Theory* 50.2 (2004), pp. 394–400. DOI: 10.1109/TIT.2003.822610.

[4] Charles H. Bennett, Gilles Brassard, and Jean-Marc Robert. "Privacy Amplification by Public Discussion". In: *SIAM Journal on Computing* 17.2 (Apr. 1988), pp. 210–229. DOI: 10.1137/0217014.

[5] Christopher C. Gerry and Peter L. Knight. *Introductory Quantum Optics*. Cambridge University Press, 2005. ISBN: 9780521527354. DOI: 10.1017/CBO9780511615099.

[6] Frédéric Grosshans et al. "Quantum key distribution using Gaussian-modulated coherent states". In: *Nature* 421 (2003), pp. 238–241. DOI: 10.1038/nature01289.

[7] Claude E. Shannon. "A Mathematical Theory of Communication". In: *Bell System Technical Journal* 27.3 (1948), pp. 379–423. DOI: 10.1002/j.1538-7305.1948.tb01338.x.

[8] A. S. Holevo. "Bounds for the Quantity of Information Transmitted by a Quantum Communication Channel". In: *Problems of Information Transmission* 9.3 (1973), pp. 177–183.

[9] John von Neumann. *Mathematische Grundlagen der Quantenmechanik*. Berlin: Springer, 1932.

[10] Christian Weedbrook et al. "Gaussian quantum information". In: *Reviews of Modern Physics* 84.2 (2012), pp. 621–669. DOI: 10.1103/RevModPhys.84.621.

[11] Theodore W. Anderson. *An Introduction to Multivariate Statistical Analysis*. 3rd ed. Hoboken, NJ: Wiley, 2003. DOI: 10.1002/0471722063.